Ali Hounain

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Work Experience

Liberty Mutual Insurance Portsmouth, NH
Technology Associate - GRS - Digital Content Solutions Jul 2026 - Present

- Engineered pre-signed **Amazon S3** payload support for an enterprise document generation API across **AWS Lambda** and **AWS Step Functions**, unblocking **4** underwriting applications whose payloads exceeded **API Gateway** limits
- Designed least-privilege access controls for application and publishing APIs across **5** environments, replacing global read/write access with granular roles and **8** new access groups
- Secured production telemetry by auditing **200**+ log statements and redacting pre-signed credentials from **AWS Lambda**, **AWS Step Functions**, and **Datadog**, preventing exposure of customer data
- Restored missing data on migrated insurance policy templates by tracing the root cause to faulty data service overrides, and deployed the fix across **5** environments

Technology Associate - GRS - Tech Modernization Jan 2026 - Jul 2026

- Led infrastructure setup for a new application tracker that improved visibility into the status of a $65 million modernization effort to modernize **273 applications**
- Engaged with stakeholders across the organization to gather requirements and feedback, ensuring the solution met their needs and providing regular project updates through demos
- Architected the backend contract for a graph-based inventory platform, defining **REST** endpoints that supported both data ingestion workflows and UI read operations
- Engineered a data ingestion pipeline that extracts, transforms and loads Excel and **Jira** data into an **AWS Neptune** graph database using **AWS Lambda**, surfacing team assignments and work status for each application

Software Engineering Intern - GDS - Messaging and Eventing Jun 2025 - Jan 2026

- Collaborated closely with the Kafka platform team to maintain self-service tools and infrastructure to support **500**+ topics using **Go** and **Spring Boot**
- Designed and deployed an event-driven solution using **Kafka** and **AWS Lambda** to automatically detect and clean up unused Kafka client applications, reducing vendor and compute costs in production
- Extended internal self-service tooling with new **REST** endpoints, enabling customers to independently diagnose and resolve infrastructure issues, reducing team service desk load

Skills

Programming Languages: Go, Java, Python, JavaScript/TypeScript, SQL

Tools/Cloud: AWS, GCP, Docker, Kubernetes, Git, Copilot

Frameworks/Libraries: React, Next.js, Node.js, Flask, Tailwind, Spring Boot

Projects

Emotion Recognition From Speech *Python, TensorFlow, Keras, Scikit-Learn* Nov 2024 - Jan 2025
https://github.com/ahounain/sentiment-analysis

- Built a customer-service-oriented tool intended to analyze a customer's emotional state based on their voice and provide companies with insights into their customer service experience
- Developed and trained a multimodal deep learning neural network for emotion recognition from audio clips in the RAVDESS dataset and achieved **90% classification accuracy**
- Implemented a dual-input neural network with Convolutional and LSTM layers to extract acoustic and temporal features
- Integrated additional features such as audio transcription and language detection

Education

University of Massachusetts Boston Boston, MA
Bachelor of Science in Computer Science Sep 2020 - Dec 2024